HDOC Day-76

Implementation of Enumerations for User Roles and Colors in C

Aim

To implement two C programs using enumerations, switch-case statements, strings, arrays and loops: one for user-role messages and another for printing twelve color names with their integer values.

Objectives

1. Define ADMIN, USER and GUEST as enumeration constants and display an appropriate welcome message.
2. Read a role from the user and handle invalid input.
3. Define an enumeration of twelve colors beginning with RED = 0, BLUE = 1 and GREEN = 2.
4. Display all twelve color names and values using a loop.

Algorithm - Program 1: User Roles

1. Define enum UserRole with ADMIN, USER and GUEST.
2. Read the role as a string.
3. Use strcmp() to map the input to an enumeration value.
4. Use switch-case to print Welcome Admin, Welcome User or Welcome Guest.
5. Display an error for an unknown role and terminate.

Algorithm - Program 2: Twelve Colors

1. Define enum Color with twelve named constants, starting at RED = 0.
2. Create a parallel array of color-name strings.
3. Iterate from RED through CYAN with a for loop.
4. Print the name and corresponding integer value for each color.

C Program - Program 1: User Roles

```
#include <stdio.h>
#include <string.h>

enum UserRole { ADMIN, USER, GUEST };

int main(void)
{
    enum UserRole role;
    char input[20];

    printf("Enter user role (ADMIN/USER/GUEST): ");
    scanf("%19s", input);

    if (strcmp(input, "ADMIN") == 0)
        role = ADMIN;
    else if (strcmp(input, "USER") == 0)
        role = USER;
    else if (strcmp(input, "GUEST") == 0)
        role = GUEST;
    else {
        printf("Invalid user role!\n");
        return 0;
    }

    switch (role) {
        case ADMIN: printf("Welcome Admin\n"); break;
        case USER:  printf("Welcome User\n"); break;
        case GUEST: printf("Welcome Guest\n"); break;
        default:    printf("Invalid user role!\n");
    }
    return 0;
}
```

C Program - Program 2: Twelve Colors

```
#include <stdio.h>

enum Color {
    RED = 0, BLUE, GREEN, YELLOW, ORANGE, PURPLE,
    PINK, BLACK, WHITE, BROWN, GREY, CYAN
};

int main(void)
{
    int i;
    const char *colorNames[] = {
        "RED", "BLUE", "GREEN", "YELLOW", "ORANGE",
        "PURPLE", "PINK", "BLACK", "WHITE", "BROWN",
        "GREY", "CYAN"
    };

    printf("Color Names and Their Integer Values:\n\n");
    for (i = RED; i <= CYAN; i++)
        printf("%s = %d\n", colorNames[i], i);

    return 0;
}
```

Explanation

Enumeration: The enum keyword defines named integer constants. Without explicit values, constants start at zero and increase by one.

User roles: ADMIN = 0, USER = 1, and GUEST = 2. The first program compares the entered string with these names, then selects the correct welcome message using switch-case.

Colors: RED is explicitly assigned 0; BLUE and GREEN automatically become 1 and 2. The other nine colors are chosen to complete the twelve-color enumeration. The loop prints each corresponding name and value.

Enumeration Values - Colors

Color	Integer Value
RED	0
BLUE	1
GREEN	2
YELLOW	3
ORANGE	4
PURPLE	5
PINK	6
BLACK	7
WHITE	8
BROWN	9
GREY	10
CYAN	11

Test Case Table

Test	Program	Input	Expected Output
1	User Roles	ADMIN	Welcome Admin
2	User Roles	USER	Welcome User
3	User Roles	GUEST	Welcome Guest
4	User Roles	MANAGER	Invalid user role!
5	Colors	No input	RED = 0, BLUE = 1, GREEN = 2, ... CYAN = 11

Sample Output - User Roles

```
Enter user role (ADMIN/USER/GUEST): GUEST
Welcome Guest

Enter user role (ADMIN/USER/GUEST): MANAGER
Invalid user role!
```

Sample Output - Twelve Colors

```
Color Names and Their Integer Values:

RED = 0
BLUE = 1
GREEN = 2
YELLOW = 3
ORANGE = 4
PURPLE = 5
PINK = 6
BLACK = 7
WHITE = 8
BROWN = 9
GREY = 10
CYAN = 11
```

Result

Both C programs demonstrate the use of enumerations. The first program prints the appropriate welcome message for ADMIN, USER or GUEST and handles invalid roles. The second prints twelve color names and their integer values using a loop, beginning with RED = 0, BLUE = 1 and GREEN = 2.